Online Case Study: Portfolio Management Simulation

BUS 104: Decision Analysis and Management Science

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Fall 2026

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Case Overview, Hard Purchase Cutoff & Key Dates

Putting Nonlinear Programming Into Action (10% of Course Grade)

Apply portfolio optimization to a live multiplayer simulation (Finsimco) across the **25 largest global companies** over **4 quarters** in **teams of 3 students**.

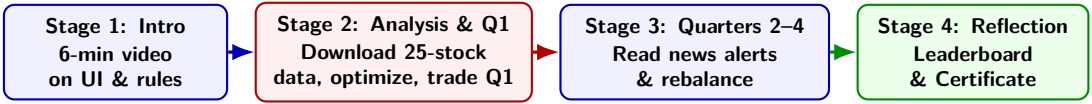
Hard Purchase Cutoff: Mon, Nov 23 (11:59 PM)

- **Course Pack:** `https://hbsp.harvard.edu/import/1456741`
- **No Exceptions:** Only students who purchase by **Mon, Nov 23 at 11:59 PM** are placed on a team. **If you miss this deadline, you cannot play and will receive 0%.**
- **Syllabus Email Rule:** You **must** register with your **exact UCR Canvas email and full name**; otherwise you will not receive credit.

Play Window & Final Deadline

- **Teams Assigned:** Teams of **3** posted on Canvas before play begins.
- **Simulation Play: Mon, Nov 30 – Fri, Dec 4** (last week of the quarter).
- **All Submissions Due: Fri, Dec 4 at 11:59 PM** (simulation completion, Canvas upload, and peer survey).

How the Simulation Works & Your Two Client Mandates



Odd Teams (1, 3, 5, . . .): Hedge Fund

- **Goal:** Beat other **Hedge Fund teams** by maximizing your portfolio **Sharpe Ratio**.
- **Tools:** May use leverage, concentrate or diversify, or hold cash.

Even Teams (2, 4, 6, . . .): Pension Fund

- **Goal:** Beat other **Pension Fund teams** by maximizing **Sharpe Ratio subject to** keeping portfolio volatility (σ_p) **below the Pension Fund median**.

Team Tip: All 3 teammates share one portfolio—designate **one member per quarter** to enter trades so you do not overwrite each other.

Syllabus Grading Scale & Peer-Adjusted Individual Scores

Syllabus Scoring Scale (10% of Grade)

Percentile	Case Score	Class Position
95–100	**100%**	Top $\sim 5\%$
90–94.9	**95%**	Next $\sim$ 5–10%
85–89.9	**91%**	Upper tier
80–84.9	**88%**	Upper-middle
75–79.9	**86%**	Middle
$>$ 0–74.9	**80%**	**Participating Floor**
No Attempt	**0% (F)**	Non-submission / Ghost

Teams are ranked by **Sharpe Ratio** within their mandate pool (Hedge vs. Hedge; Pension vs. Pension). Valid participation guarantees a **team score** $\geq 80\%$.

Peer Survey & Downward Adjustment Below 80%

- Purchasing the case alone **does not** earn 80%.
- Every student submits a **Canvas Peer Survey** splitting **100 pts** across their 2 teammates (missing survey = 50/50 split for teammates).
- Non-working students have their score **adjusted downward below 80%** (down to **0%**).

Team-of-3 Math Example (Team Score = 80%)

Team earns **80%**. **A & B** work; **C** ghosts or under-contributes:

- **A & B** ($\sim 50/50$ peer share): earn $\geq 80\%$.
- **C gets** $0/100$ from both $\implies 80\% \times 0 = \mathbf{0\%}$.
- **C gets** $20/100$ (vs. 50 expected) $\implies 80\% \times \frac{20}{50} = \mathbf{32\%}$.

Peer Survey Rules: Self-Rating & Conflicting Reports

1. You Cannot Rate Yourself

- In the Canvas Peer Survey, you **exclude yourself** and divide **100 points** strictly between your **2 teammates** (e.g., $50/50$ for equal work).
- Nobody can award themselves 100 points.
- If you skip the survey, your unsubmitted ballot defaults to a **50/50 split** for your teammates (so you cannot hurt them by forgetting).

2. Consensus Rule

- Automatic downward adjustment applies when **both teammates** independently report low/zero effort (e.g., $< 20/100$ each).

3. What If Teammates Submit Wildly Different Reports?

Suppose **A** says **B** did nothing, **B** says **A** did nothing, and **C** says everyone worked equally:

- **One teammate alone cannot tank your grade.** Conflicting surveys do **not** trigger automatic penalties.
- Instead, conflicting reports trigger an **Objective Tie-Breaker Audit** using:
  1. **Finsimco *Live Monitoring* logs** (login timestamps and Q1–Q4 activity).
  2. **Canvas Portfolio Verification** (your individual Q1–Q4 trade answers).
  3. **Timestamped Proof** (dated group chat messages, Colab/Excel edits, or outreach sent *before* the deadline).

Required Deliverables (Due Friday, Dec 4 at 11:59 PM)

All 3 Items Must Be Submitted Individually on Canvas by Fri, Dec 4 (11:59 PM)

1. **Finsimco Certificate of Completion (PDF/Screenshot):** Downloaded from Stage 4 (*Reflection*).
2. **Optimization File (`.ipynb` or Excel) + Brief Summary:** Your Q1 model plus 2–3 sentences on your top Q1 stocks and Q2–Q4 rebalancing.
3. **Canvas Peer Evaluation Survey:** Split 100 points across your 2 teammates (required to claim your team's score).

48-Hour No-Show Rule (Protecting Active Students)

- If a teammate does not respond within **48 hours** of team assignment (by **Wed, Dec 2**), screenshot your outreach message and **proceed without them**.
- A team of 2 (or 1) can complete every stage in Finsimco. Active students are **never** penalized for a ghosting teammate; non-participants are verified via Finsimco *Live Monitoring* and adjusted down (to 0% if zero work).

Quick Python Starter: Scaling Lecture Code to 25 Stocks

In **Google Colab**, compute `mu` and `Sigma` directly from Finsimco's downloadable Excel file:

```
import pandas as pd, numpy as np
from scipy.optimize import minimize, LinearConstraint

prices = pd.read_excel("historical_data.xlsx", index_col=0)
returns = prices.pct_change().dropna()
mu = returns.mean().values * 252          # Annualized expected returns (25 stocks)
Sigma = returns.cov().values * 252        # Annualized 25x25 covariance matrix

def neg_sharpe(w, rf=0.03):               # Minimize NEGATIVE Sharpe Ratio
    return -(np.dot(mu, w) - rf) / np.sqrt(np.dot(w, np.dot(Sigma, w)))

n = len(mu)
res = minimize(neg_sharpe, x0=np.ones(n)/n, bounds=[(0, 1)]*n,
               constraints=[LinearConstraint(np.ones(n), 1.0, 1.0)])
```

Tip for Even Teams (Pension Funds): Also check your portfolio volatility $\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}$ so it stays safely below the Pension Fund median.

Summary Schedule & Action Checklist

Date & Time	Milestone	What You Must Do
Mon, Nov 23 (11:59 PM)	**HARD PURCHASE CUTOFF**	Buy course pack at `hbsp.harvard.edu/import/1456741` using your **exact Canvas email & full name**. Miss = 0%.
Before Mon, Nov 30	**Teams of 3 Assigned**	Check Canvas for your Team # (Odd = Hedge Fund, Even = Pension Fund) and message your teammates.
Mon, Nov 30 – Fri, Dec 4	**Play Window (Q1–Q4)**	Watch Intro video, optimize Q1 portfolio, and complete Q2–Q4 rebalancing trades. (48-hr No-Show Rule applies by Wed, Dec 2.)
Fri, Dec 4 (11:59 PM)	**ALL SUBMISSIONS DUE**	Submit (1) Finsimco Certificate, (2) Model + summary, and (3) Canvas Peer Evaluation Survey.